

Octupole-driven spin-transfer torque switching of all-antiferromagnetic tunnel junctions

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Magnetic tunnel junctions (MTJs) based on ferromagnets are canonical devices in spintronics, with wide-ranging applications in data storage, computing, and sensing. They simultaneously exhibit mechanisms for electrical detection of magnetic order through the tunneling magnetoresistance (TMR) effect, and reciprocally, for controlling ferromagnetic order by electric currents through spin-transfer torque (STT). It was long assumed that neither of these effects could be sizeable in tunnel junctions made from antiferromagnetic materials, since they exhibit no net magnetization. Recently, it was shown that all-antiferromagnetic tunnel junctions (AATJs) based on chiral antiferromagnets do exhibit TMR due to their non-relativistic momentum-dependent spin polarization and cluster magnetic octupole moment (CMO). However, the reciprocal effect, i.e., the antiferromagnetic counterpart of STT, has been assumed non-existent due to the total electric current being spin-neutral. Here, in contrast to this common expectation, we report nanoscale AATJs exhibiting this reciprocal effect, which we term octupole-driven spin-transfer torque (OTT). We demonstrate current-induced OTT switching of PtMn₃|MgO|PtMn₃ AATJs, fabricated on a thermally oxidized silicon substrate, exhibiting a TMR value of 363% at room temperature and switching current densities of the order of 10 MA/cm². Our theoretical modeling explains the origin of OTT in terms of the imbalance between intra- and inter-sublattice spin currents across the AATJ, and equivalently, in terms of the non-zero net cluster octupole polarization of each PtMn₃ layer. This work establishes a new materials platform for antiferromagnetic spintronics and provides a pathway towards deeply scaled magnetic memory and room-temperature terahertz technologies.

Index Terms— magnetic tunnel junction, spin-transfer torque, noncollinear antiferromagnet, cluster magnetic octupole

I. INTRODUCTION

Antiferromagnetic materials have emerged as promising candidates for spintronics due to their potential for higher-density integration, immunity to external fields, possibility of manipulation by spin-orbit torque and potential ultrafast operation [1]. A key goal is to use these materials as the active element of the magnetic tunnel junction (MTJ) structure, replacing the role of ferromagnets while retaining the same read and write functions [2]. Recently, all-antiferromagnetic tunnel junctions (AATJs) were demonstrated to exhibit a sizable room-temperature tunneling magnetoresistance (TMR) by employing identical noncollinear antiferromagnets for both electrodes [3–5]. Despite their vanishing net magnetization, they exhibit a non-relativistic momentum-dependent spin polarization or CMO polarization which reverses with opposite spin configurations, leading to a finite TMR value. However, the reciprocal effect, analogous to spin-transfer torque (STT) in ferromagnet-based MTJs and its utilization for switching of antiferromagnetic layers in AATJs have remained elusive.

Here, we experimentally demonstrate octupole-driven spin-transfer torque (OTT)-induced switching of the cluster magnetic octupole (CMO) by flowing a vertical current through nanoscale PtMn₃|MgO|PtMn₃ AATJs [6]. The devices are grown by sputter deposition on thermally oxidized silicon wafers, followed by patterning into pillars with diameters ranging from 50 to 200 nm. We find that current flowing through the junction drives the switching of the magnetic octupole moment without an external magnetic field at a switching current density (J) of the order of 10 MA/cm². This switching results in a resistance variation of the device, with a TMR ratio up to 363% at room temperature. Our density functional theory (DFT) calculations show that imbalance

between intra-sublattice and inter-sublattice torque manifests itself as a net staggered spin-transfer torque acting separately upon each sublattice. Our results provide a foundation for implementing all-antiferromagnetic magnetoresistive random-access memory.

II. EXPERIMENTAL OBSERVATIONS OF OTT-INDUCED SWITCHING

We first deposited Pt (5 nm) / PtMn₃ (7 nm) / MgO (2 nm) / PtMn₃ (10 nm) / Pt (5 nm) / Ru (10 nm) stacks at room temperature on thermally oxidized 150 mm silicon wafer, which were subsequently patterned into nanometer-sized circular pillars ranging from 50 to 200 nm. Both top and bottom PtMn₃ exhibits predominant (111) orientation, confirmed by scanning transmission electron microscopy. Two-terminal current-induced switching measurements at room temperature shows bidirectional switching with a sharp transition at the current density, J of ~ 11 MA/cm² with a TMR value of 363%. Statistical distribution of TMR within 50 devices including all sizes of pillars indicate that current-induced switching is consistently present in our devices. We also carried out current-induced switching measurements at different temperatures up to 400 K. We found that the current-induced switching and TMR both vanished at temperatures near 360 K, consistent with PtMn₃ undergoing a magnetic phase transition from a noncollinear to a collinear phase at elevated temperature and thus further supports the fact that the switching of our AATJs arises from the noncollinear spin configuration of the PtMn₃ layers.

III. MICROSCOPIC MECHANISM OF OTT

We next present a microscopic picture for the observed spin torque in terms of the sublattice-resolved spin polarizations. The atom- and site-spin-projected conduction channels exhibit

sublattice-dependent anisotropic distribution across the Brillouin zone. The anisotropic distribution indicates that the electrons preferentially retain their sublattice identity and spin as they tunnel from one PtMn₃ layer to the other. As the total spin polarization of the inter-sublattice current is in the opposite direction of the intra-sublattice polarization, the total tunneling current to each sublattice carries a net spin polarization and induces an OTT to the other PtMn₃ layer. This torque resembles a sublattice-dependent STT and is a direct consequence of the octupolar order. Simulations of the dynamics using a three-sublattice Landau-Lifshitz-Gilbert (LLG) model demonstrate that this OTT mechanism can indeed give rise to deterministic and directional current-induced switching.

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